

MAN-MADE CELLULOSIC FIBRES

VISCOSE RAYON

Definition

The textile fibre products identification Act (TFPIA) defines rayon as "a

Manufactured fibre composed of regenerated cellulose, as well as manufactured fibres composed of regenerated cellulose in which substituted have replaced more than 15% of the hydrogen of the hydroxyl radicals.

Manufacturing process

- The principal raw materials or viscose rayon is a refined grade of pulp derived from wood.
- Cellulose is extracted from wood, purified and formed into thin sheets that are 2 feet (0.61m) square.
- The sheets of cellulose are steeped in a 17.5% solution of sodium hydroxide (NaOH) until the cellulose is converted to alkali cellulose.
- After steeping the alkali cellulose sheets are shredded into a crumb.
- The crumb is then aged for varying periods depending on type of rayon.
- The aging process involves a chemical reaction between the alkali cellulose and oxygen in the air.
- After aging, the alkali cellulose are reacted with carbon disulfide, to produce sodium cellulose xanthate (bright orange in colour).
- The xanthate crumb is dissolved in dilute sodium hydroxide (5-8% NaOH) and form honey coloured liquid.

- This solution is aged until it reaches the correct degree of coagulability with respect to its spinning ability for specified spinning conditions.
- The name viscose is taken from the viscous liquid solution formed in this manufacturing step.
- The viscose is filtered and de-aerated, delivered to the spinning machines and forced by individual pumps for each spinnerette into an acid bath.
- The spin bath for viscose rayon is composed of sulphuric acid, sodium sulfate, and water and certain additives such as glucose and zinc sulfate.
Sodium sulfate is the primary coagulation medium.
- Water is the basic solvent for the spin bath.
- The filaments are thoroughly washed after regeneration to remove any impurities that might adhere to the fibre.
- The formed fibres are withdrawn from the spun bath and either combined directly into yarns or cut into short lengths for spinning into yarns by one of the methods used for cotton or wool fibres.

Chemical Reactions in Viscose Manufacture

1. Cellulose + caustic soda = cellulose + water

$$(C_6H_{10}O_5)_n + nNaOH \rightarrow (C_6H_9O_5Na)_n + nH_2O$$
2. Soda cellulose + carbondisulfide = sodium cellulose xanthate

$$(C_6H_9O_4C_2Na)_n + n/2H_2SO_4 \rightarrow (C_6H_9O_4OCS_2Na)_2$$
3. Aging period
4. Sodium cellulose xanthate + sulphuric acid = cellulose (viscose type)

FIBER PROPERTIES

Microscopic Properties

Longitudinally viscose rayon has a relatively uniform diameter and may appear to have striations. A delustrated fiber will have a grainy, pitted appearance, while brighter fibers appear transparent. The cross sectional view of viscose rayon is highly irregular and crenulated.

Physical Properties

a) Shape and Appearance

Rayon fibers can be produced in any length desired. The length and denier are determined by the manufacture for the desired end use. They are available in both filament and staple lengths. Diameter varies from 12 to 40 microns.

b) Luster

Luster can be controlled. It varies from bright to dull.

c) Tenacity (strength)

It varies from 2.4 to 5g/d. Viscose rayon is weaker when wet than when dry, hence should be handled carefully when laundering.

d) Elastic Recovery and Elongation

The processing of viscose rayon is modified to provide fibers that have low extensibility and high tenacity, and also fibers with high elongation and low tenacity.

e) Density

The density of rayon is about 1.5g/cm³. This is similar to natural cellulosic fibers and this means that fabrics of rayon made the same as those for cotton will be similar in weight.

f) Resiliency

The resiliency of viscose rayon is relatively low.

g) Moisture Regain

The moisture regain of viscose rayon is slightly higher than that of natural cellulosic fibers. This makes the fiber dye and finish much better than cotton. This is in part due to the lower degree of polymerization, i.e. there are more amorphous areas into which water, dyes and finishes can be absorbed.

h) Dimensional Stability

Regular rayons are subject to easy stretching during yarn and fabric manufacture, followed by relaxation shrinkage after laundering.

Finishes can be used on viscose rayon to control dimensional stability.

Thermal Properties

Viscose rayon fibers burn like other cellulosic fibers with a yellow flame and relatively rapidly. A small amount of light grey or off-white fluffy residue is left. When the flame is extinguished, there may be an after glow.

Ironing temperatures similar to those used for cotton are acceptable. Ironing temperatures should not exceed 140 degrees Fahrenheit.

Chemical Properties

a) Effect of Alkalis

Similar to cotton but the effect occurs at a faster rate than they would with cotton.

b) Effect of Acids

Hot and cold concentrated inorganic acids cause rayon fibers to disintegrate. Hot dilute acids result in fiber breakdown, but cold dilute acids have little or no effect.

c) Effect of Organic solvents

Viscose rayon has good resistance to dry cleaning solvents and stain removal agents. Many other organic compounds have little or no harmful effect on the fibres.

Effect of Sunlight Age and Miscellaneous Factors

Rayon fibers undergo deterioration when exposed to ultra violet rays of the sun for long periods of time. They have good resistance to ageing and can last for many years when properly stored. They should be stored clean and dry in relatively dark areas.

Static Charges

Rayon fibers do not build up static charges.

Biological Properties

Viscose rayon is resistant to all insects. Silverfish, which may attack cellulose may damage rayon fabrics. Mildew will destroy all rayons.

Use and Care of Viscose Rayon

Used for apparel and home furnishing fabrics. It makes a good choice for blending with other fibres, such as polyester, acrylic, cotton and wool.

Viscose rayon fabrics should be handled with care during laundering. It is not affected by most detergents and other laundry agents such as starches, fabric softeners and water softeners.

ACETATE AND TRIACETATE

Definition

Acetate is defined, by TEPIA, as a manufactured fibre in which the fibre forming substances are cellulose acetate.

Manufacture of acetate rayon

The raw materials for manufacturing regular or secondary acetate include cellulose, acetic acid and acetic anhydride plus sulphuric acids as a catalyst.

Cellulose from either wood pulp or cotton linters may be used but wood pulp is the more common source.

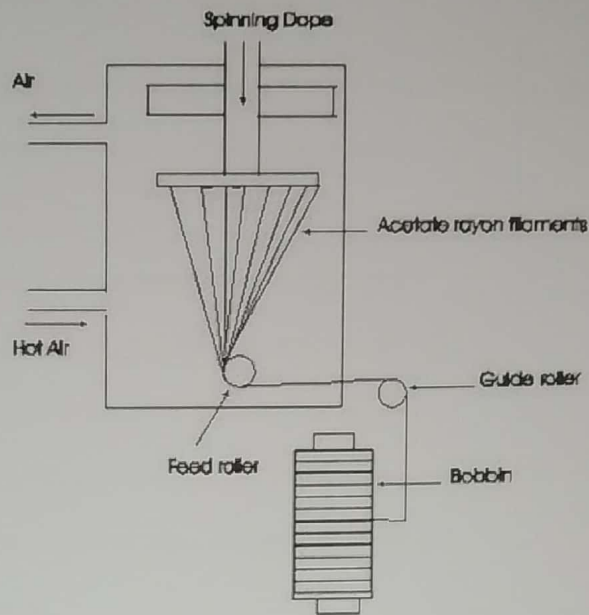
1. The cellulose is purified, bleached and shredded and then it is fed into tanks where it is thoroughly mixed with glacial acetic acid.
2. Pretreated cellulose is transferred to kneading machines called acetyletor, where acetic anhydride is added. During this step the pre-treated cellulose assumes a liquid form as a new chemical compound, cellulose acetate. At this point, the cellulose acetate is a clear liquid called acid dope.
3. It is emptied into special storage tanks where it is aged or ripened. Water is added to the solution to reduce the acid concentrate from 100% - 95%.
4. After ripening the secondary acetate is mixed with water and the acetate precipitates out in the form of small flakes. During ripening hydrolysis occurs which removes some of the acetyl groups from various parts of the cellulose molecule, replacing it with a hydroxide radical and producing regular acetate acetyl radical. The flakes are washed free of excess acid and dried. During precipitation and washing excess acetic acid sulphuric acid are recovered for reuse to reduce the cost of manufacture as well as pollution of the environment.
5. The cellulose acetate flakes is dissolved in acetone to form the spinning dope.
6. The dope is forced through fibres to remove any undissolved acetate and impurities, then forced through spinnerette into a warm air chamber. Acetone evaporates and is recovered. The acetate coagulate as it falls through the warm air chamber. The filament then twisted together to form yarn. (Dry spinning).

- (a) Cellulose + acetic anhydride = cellulose triacetate
- (b) Cellulose + acetate + water = secondary acetate

TRIACETATE

This is manufactured from the same raw materials secondary or regular acetate. The process is identical with that used for regular acetate up to the ripening step which is omitted for triacetate. This eliminates any hydrolysis action and leaves triacetate flakes. The flakes are dried then dissolved in methylene chloride instead of acetone. The solution is forced through the spinnerette into a warm air chamber.

Flow Diagram Showing Processes in Acetate Manufacture



FIBRE PROPERTIES OF ACETATE AND TRIACETATE

Microscopic Properties

- The longitudinal view shows acetate as uniform in width with several lines parallel to the length of the fibre.
- The cross section of acetate shows that fibres are lobed in outline, with irregular curves but there are no sharp points (such as in viscose).
- Triacetate is very similar in microscopic appearance to acetate. The longitudinal view shows, in addition to striations a speckled or pitted look associated with delustering.
- The cross section is lobed and similar to acetate. A positive distinction between acetate and triacetate cannot be made by microscopic examination.

Physical Properties

- (a) Shape and appearance
 - Acetate and triacetate can be made to any length in a variety of diameters from very fine to heavy fibres.

(b) Lusture

- The lusture of acetate and triacetates can be controlled therefore fibres can be obtained that are very dull, very bright and very shinny, or at any stage between.

(c) Tenacity

- Acetate has a dry tenacity of 1.2-1.4g/d. wet strength is 0.8-1.0g/d
- Triacetate has a dry tenacity of 1.1-1.4g/d and wet tenacity of 0.8-1.0g/d.
- The strength of both acetate and triacetate is relatively low when compared with that of many fibres. However, the fibres have some desirable properties that make it attractive for use in selected types of apparel and home.

(d) Elongation and Elastic Recovery

- Acetate and triacetate have good elastic recovery and elongation.
- If stretched more than 1%; acetate undergoes permanent deformation and will not return to its original size.
- The acetate has slightly less recovery 2% extension.

(e) Resiliency

- Regular acetate does not have good recovery from crushing or wrinkling. Its resiliency is low.
- Triacetate possess good resiliency, and it is properly heat-treated, Wrinkles hang out quickly and relatively completely.

(d) Density

- The density of both secondary acetate and triacetate is 1.32.
- Both are higher than regular cellulose.

(g) Moisture Regain

- Standard moisture regain for acetate is 65% and for triacetate is 3.2-3.5%.
- At saturation, acetate will absorb up to 14% moisture whereas triacetate will absorb 9%.

(h) **Dimensional Stability**

- both acetate and triacetate fibres are comparatively resistant to stretch or shrinkage, except when relaxation shrinkage might occur because tensions applied when constructing fabrics.
- The fibres may shrink when exposed to high temperatures.

Thermal Properties

- Regular acetate is easily softened by temperatures above 177o C (380o F).
- The fibre melts and burns evenly forming a hard black bead ash or residue.
- It gives off a chemical odour or smell similar to hot vinegar.
- The fibre should be ironed at low to medium temperatures setting.
- Triacetate melts and burns in the same manner as regular acetate.
- Triacetate can be heat treated to withstand high temperatures (212o C) 450o F without damage e.g. heat setting pleats.

Chemical Properties

Alkalis

- Dilute alkalis have little effect on either acetate or triacetate.
- Concentrated alkalis damage them.

Acids

- Concentrated acids weaken the fabrics in most cases complete fibre degradation.
- Hot acids both dilute and concentrated may cause decomposition or loss of strength.
- Cold dilute acids may weaken the fibre if exposure is prolonged.

Effect of organic solvents

- Most petroleum solvents used in dry cleaning do not damage acetate fibres.
- However solvents such as acetone phenol chloroform will destroy the fibre.

- Nail varnish removers destroy acetate.
- It should be handled carefully when removing stains because some stain removers contain chemicals that might damage the fibres.

Effect of age, sunlight and miscellaneous factors

- Acetate loses strength and may split after prolonged exposure to sunlight.
- Over time acetate will become weaker for aging.
- Triacetate has excellent stability to aging and is more resistant to sunlight than acetate silk or nylon.

Static charges

- All acetates develop static charge especially when the air is very dry.

Biological properties

- Fungi such as mildew and bacteria may discolour acetate and triacetate. Acetate may be weakened by mildew but triacetate retains its strength.
- Insects – moth and other household pests do not damage acetate fibres.
- However, silverfish may attack any sizing on acetate and triacetate.

USE AND CARE OF ACETATE AND TRIACETATE FABRICS

- Other apparel including suits and dresses jackets, coat linings, draperies, upholstery, and pillow covers.
- Triacetate - Makes outstanding knits, makes home furnishing, domestic and apparel fabrics just like acetate.
- Triacetate can be heat-set into pleats
- Acetate is used as the filter media for cigarettes.